

XAI 509 Course Project

Wav2Vec2 CTC Fine-Tuning with Output-Distribution Regularization

LibriSpeech 1-hour Fine-tuning · Maximum Entropy vs Label Smoothing

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June 11, 2026

Outline



- Background & Problem

- Two Regularization Methods

- Experimental Setup

- Main Results

- MaxEnt + LS Sweeps & Analysis

- Summary

Project Overview

| Wav2Vec2 CTC Fine-Tuning + Regularization on LibriSpeech 1h

Goal

Fine-tune **Wav2Vec2** with the **CTC loss** on a small (1-hour) LibriSpeech subset and report WER on test-clean / test-other.

Core Idea

Small fine-tuning data → **over-confidence**. Compare two output-distribution regularizers: **Maximum Entropy** and **Label Smoothing**.

Key Finding

Both regularizers improve over baseline. **Label Smoothing ($\alpha=0.10$) achieves the best WER** — clean **19.51 %** (−1.65 pp), other **28.30 %** (−1.87 pp), outperforming both baseline CTC and Maximum Entropy training.

Scope / Limitation

Only $\alpha = 0.10$ swept for label smoothing; MaxEnt has 5-point λ sweep. *Single-seed experiments.*

Background

| Wav2Vec 2.0 + CTC for ASR

Wav2Vec 2.0

- Self-supervised speech representation
- Pretrained on 53k hours of unlabeled audio
- CNN feature encoder → Transformer context
- Fine-tunable for downstream ASR

CTC Loss

- No frame-level alignment required
- Marginalizes over all valid alignments
- $WER = (S + I + D) / N$
- Lower WER = better recognition

Why this project?

Fine-tuning Wav2Vec2 on just **1 hour** of speech is challenging — the model can become **over-confident on training data**, producing sharp output distributions that fail to generalize to **test-clean / test-other**.

Method

Two Regularization Strategies

Both methods penalize sharp output distributions, but apply different weighting schemes.

Maximum Entropy

$$\text{Loss} = \text{CTC} - \lambda \cdot H(p)$$

$$H(p) = -\sum p_v \log p_v$$

Properties:

- Weights penalty by predicted probability p
- Pushes confident tokens toward uniform first
- Listed in lecture 8 "Topics You Can Try"

Label Smoothing

$$\text{Loss} = (1-\alpha) \cdot \text{CTC} + \alpha \cdot \text{LS}(p)$$

$$\text{LS}(p) = -(1/V) \sum \log p_v$$

Properties:

- Weights penalty uniformly across vocab ($1/V$)
- Equivalent to $\text{KL}(\text{uniform} \parallel p)$ up to constant
- Reported as part of lab's MoChA paper

→ Closely related (Pereyra et al. 2017) but mathematically distinct.

Experiment

Setup

Pretrained model	facebook/wav2vec2-base (95M params)
Train set	LibriLight 1-hour subset (WebDataset, 5 shards)
Eval sets	test-clean (2620 utt) / test-other (2939 utt)
Training steps	2000 (~222 epochs on 1h set)
Optimizer / LR	AdamW, lr=1e-4, warmup 500 steps
Batch	16 × grad_accum 2 = effective 32, fp16
Frozen modules	CNN feature encoder
Hardware	RTX 3090 (24 GB), ~35 min per run

Each variant (baseline / MaxEnt λ / Label Smoothing α) trained with identical settings for fair comparison.

Results

| Main Comparison — Word Error Rate (%)

Method	test-clean	test-other	Δ vs Base
Baseline (CTC only)	21.16	30.17	—
Maximum Entropy (best, $\lambda = 0.15$)	19.88	28.63	-1.54
Label Smoothing (best, $\alpha = 0.10$)	19.51	28.30	-1.87

Takeaways

- Both regularizers beat baseline by **> 1.5 pp** on both splits (best operating points).
- Label Smoothing ($\alpha=0.10$) is the overall winner — **0.37 pp (clean) / 0.33 pp (other) ahead of MaxEnt**.
- Each curve was tuned by a full sweep (next slides): **LS peaks sharply at 0.10, MaxEnt plateaus at 0.15-0.20**.

Results

Maximum Entropy — λ Sweep

λ	test-clean	test-other
0.00 (baseline)	21.16	30.17
0.01	20.60	29.67
0.03	20.79	29.65
0.05	21.21	30.46
0.07	20.93	29.80
0.10	19.93	28.80
0.15	19.88	28.63
0.20	20.09	28.60

Observations

- **Small λ (0.01-0.03)** gives modest improvement
- **$\lambda = 0.05$** dips — a noisy bump
- **$\lambda = 0.10$** strong, but not the best
- **$\lambda = 0.15-0.20$** lowest WER (flat optimum)

Best operating point:

$\lambda = 0.15 \rightarrow 19.88 / 28.63$

Broad optimum $\rightarrow$ robust to λ choice.

Results

Label Smoothing — α Sweep

α	test-clean	test-other
0.00 (baseline)	21.16	30.17
0.01	20.90	30.15
0.03	20.71	29.98
0.05	21.43	30.09
0.07	20.91	29.75
0.10	19.51	28.30
0.15	20.16	29.39
0.20	20.69	29.43

Observations

- $\alpha \leq 0.07$ only marginal gains
- $\alpha = 0.10$ sharp minimum — best by far
- $\alpha \geq 0.15$ regresses again
- **Sharp U-curve** → sensitive to α

Best operating point:

$\alpha = 0.10 \rightarrow 19.51 / 28.30$

Overall best of all configs.

Analysis

Why Label Smoothing > MaxEnt

Same family, different weighting:

- **MaxEnt:** $H(p) = -\sum p_v \log p_v \rightarrow$ weights by p_v — already-confident tokens get pushed first
- **LS:** $LS(p) = -(1/V) \sum \log p_v \rightarrow$ uniform weight across vocab — all tokens regularized equally

Qualitative example (test-clean utterance 1)

REF: ... THEORY OF MEMORY MUST ARRIVE AT ... VIRTUE ... SINGLED OUT ...

Baseline: ... THEORY OF MEMERY MUST ARIVE AT ... VERTUE ... SINGLEDOUT ...

LS $\alpha=0.10$: ... THEORY OF MEMERY MUST ARRIVE AT ... VIRTUE ... SINGLED OUT ...

→ Spelling errors (ARIVE, VERTUE) and spacing (SINGLEDOUT) corrected by Label Smoothing.

Summary

01

Two regularizers, fully swept

Maximum Entropy (λ , 8 points) and Label Smoothing (α , 7 points), both vs the CTC baseline, all under identical training settings.

02

Label Smoothing wins

Best $\alpha=0.10 \rightarrow 19.51 / 28.30$. Beats baseline by $-1.65 / -1.87$ pp and the best MaxEnt ($\lambda=0.15$) by $-0.37 / -0.33$ pp.

03

Different curve shapes

LS has a sharp optimum at $\alpha=0.10$ (sensitive); MaxEnt plateaus over $\lambda=0.15-0.20$ (robust but lower peak). Gains largest on test-other.

Future: combine LS + MaxEnt, SpecAugment data augmentation, LM shallow fusion, multi-seed runs.

Thank You

Q & A

Code & results: [/run/models/](#), [/run/results/](#)